$$= \overline{(x_1 + x_2) + i(y_1 + y_2)} = (x_1 + x_2) - i(y_1 + y_2)$$

$$= (x_1 - iy_1) + (x_2 - iy_2)$$

$$= \overline{z_1} + \overline{z_2}\ .$$

It can be generalized by means of mathematical induction to sums involving any finite number of terms: $\overline{z_1 + z_2 + z_3 + \cdots z_n} = \overline{z_1} + \overline{z_2} + \overline{z_3} + \cdots + \overline{z_n}$.

Property

$\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$ where $x_1, x_2, y_1,$ and $y_2 \in \mathbb{R}$

Proof

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$.

Then, $z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$.

Therefore, $\overline{z_1 z_2} = \overline{(x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)} = (x_1x_2 - y_1y_2) - i(x_1y_2 + x_2y_1)$,

and $\overline{z}_1\overline{z}_2 = (x_1 - iy_1)(x_2 - iy_2) = (x_1x_2 - y_1y_2) - i(x_1y_2 + x_2y_1)$.

Therefore, $\overline{z_1 z_2} = \overline{z}_1\overline{z}_2$.

Property

A complex number z is purely imaginary if and only if $z = -\overline{z}$

Proof

Let $z = x + iy$. Then by definition $\overline{z} = x - iy$

Therefore, $z = -\overline{z}$

$\Leftrightarrow \quad x + iy = -(x - iy)$

$\Leftrightarrow \quad 2x = 0 \Leftrightarrow x = 0$

$\Leftrightarrow z$ is purely imaginary.

Similarly, we can verify the other properties of conjugate of complex numbers.

Example 2.3

Write $\dfrac{3+4i}{5-12i}$ in the $x + iy$ form, hence find its real and imaginary parts.

Solution

To find the real and imaginary parts of $\dfrac{3+4i}{5-12i}$, first it should be expressed in the rectangular form $x + iy$. To simplify the quotient of two complex numbers, multiply the numerator and denominator by the conjugate of the denominator to eliminate i in the denominator.

$$\frac{3+4i}{5-12i} = \frac{(3+4i)(5+12i)}{(5-12i)(5+12i)}$$

$$= \frac{(15-48)+(20+36)i}{5^2+12^2}$$

$$= \frac{-33+56i}{169} = -\frac{33}{169} + i\frac{56}{169}\ .$$

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